



50 Core Java Interview Questions (Before OOPs Concepts)

 **Level:** Beginner to Intermediate

 **Scope:** Everything before Classes, Objects, Inheritance, etc.

♦ 1–10: Java Basics & Environment

1. What is Java?
2. Explain the features of Java (platform-independent, secure, etc.).
3. What is the difference between JDK, JRE, and JVM?
4. What is bytecode in Java?
5. What is the role of the `main()` method in Java?
6. Why is Java called platform-independent?
7. How is Java different from C++?
8. What is the use of the `javac` and `java` commands?
9. What are comments in Java? List all types.
10. What is the default package in Java?

◆ 11–20: Data Types, Variables, and Keywords

11. What are the different data types in Java?
 12. What is the difference between primitive and non-primitive data types?
 13. What is the default value of an uninitialized variable?
 14. How is memory allocated for variables in Java?
 15. What is type casting? Explain implicit and explicit casting.
 16. What is the size and range of the `int` data type in Java?
 17. What is the use of the `final` keyword?
 18. What is the difference between `var`, `int`, and `Integer` in Java (Java 10+)?
 19. Can we declare multiple variables in a single line in Java?
 20. What are reserved keywords in Java? List any 5.
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◆ 21–30: Operators and Expressions

21. What are the types of operators in Java?
 22. Explain the difference between `==` and `.equals()` in Java.
 23. What is the output of `5 + 3 * 2` in Java? Why?
 24. What is a ternary operator? Provide an example.
 25. Explain increment (`++`) and decrement (`--`) operators with examples.
 26. What are bitwise operators?
 27. How do logical operators (`&&`, `||`) work in Java?
 28. What is short-circuit evaluation in Java?
 29. What are assignment operators in Java?
 30. What is the difference between `&` and `&&`?
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◆ 31–40: Control Flow – if, switch, loops

31. What is the difference between `if` and `switch` statements?
32. When would you use `if-else` vs `switch-case`?
33. Can `switch` use `String` values in Java?

34. What is the syntax of a `while` loop?
 35. What is the difference between `for`, `while`, and `do-while`?
 36. How does the `break` statement work in a loop?
 37. What is the use of the `continue` keyword?
 38. What is a nested loop? Provide an example.
 39. What are labeled loops in Java?
 40. How do you print patterns using loops in Java?
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◆ **41–50: Arrays, Strings, and Methods (Pre-OOP)**

41. What is an array in Java? How is it declared?
42. What is the default value of an array element in Java?
43. Can an array size be changed once declared? Why?
44. What is the difference between 1D and 2D arrays in Java?
45. What are the different ways to declare and initialize an array?
46. What is a String in Java? How is it different from a `char[]` array?

47. Explain immutability of Strings in Java.
48. What is the use of `StringBuilder` and `StringBuffer`?
49. How are methods defined in Java?
50. What is method overloading? (Introduced briefly before full OOP)